Lecture 7

Compiler Design CSC-305

Awanish Pandey

Department of Computer Science and Engineering
Indian Institute of Technology
Roorkee

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4. Remove $x_1 \ldots x_i$ from input; go to (2)

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- Regular expressions provide a concise and useful notation for string patterns
- Good algorithms require a single pass over the input

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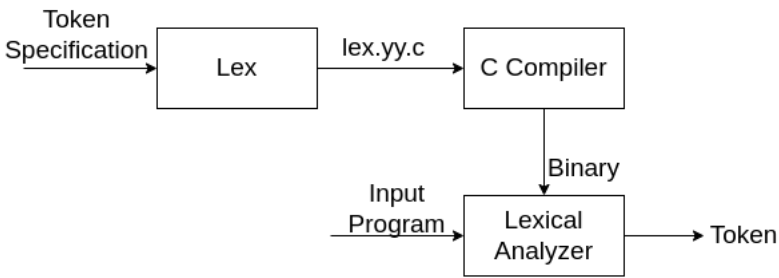

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transition rules
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auxiliary functions

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  %{
  include < *math.h* >
  int count;
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Transition Rule

Pattern $\{Action\}$

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C Code, going directly into `lex.yy.c`

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- Emit code driven by the DFA tables

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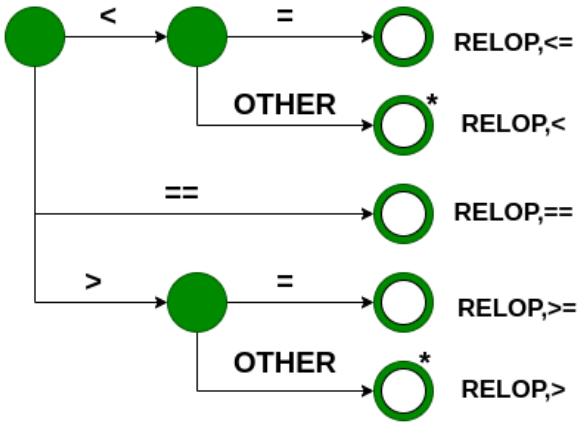
- Lexical definitions consist of **regular definitions, priority rules**, and **maximal munch principle**
- Construct an analyzer that will return ⟨`token, lexeme`⟩ pairs

Transition Diagram for Relational Operator



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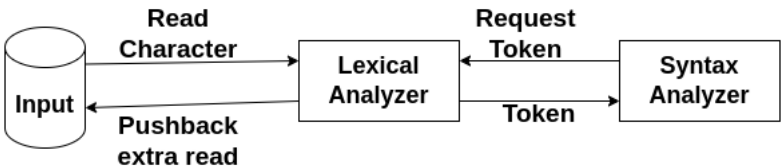
- This can be implemented using **lots of** `switch-cases` in C programming language.

Interface to other passes



Syntax Analysis

Compiler Design CSC-305

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- To check whether a variable has been declared before use ✗
- To check whether a variable has been initialized ✗
- These issues will be handled in semantic analysis

Derivation

Does $9 - 5 + 2$ belong to the following grammar?

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\begin{aligned}
\textit{list} &\rightarrow \textit{list} + \textit{digit} \\
&\quad | \textit{list} - \textit{digit} \\
&\quad | \textit{digit} \\
\textit{digit} &\rightarrow 0|1|2 \ldots |9
\end{aligned}
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- If $S \Rightarrow^+ \alpha$ where α is a string of terminals and non-terminals of G then we say that α is a **sentential** form of G.

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- Every leftmost step can be written as $wA\gamma \Rightarrow^{lm*} w\delta\gamma$ where w is a string of terminals and $A \rightarrow \delta$ is a production.

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- If $S \Rightarrow^+ \alpha$ where α is a string of terminals and non-terminals of G then we say that α is a **sentential** form of G.
- If in a sentential form only the leftmost non terminal is replaced then it becomes leftmost derivation
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- An ambiguous grammar is one that produces more than one leftmost/rightmost derivation of a sentence

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- If A is a non-terminal labeling an internal node and $x_1, x_2, \ldots x_n$ are labels of the children of that node, then $A \rightarrow x_1 x_2 \ldots x_n$ is a production

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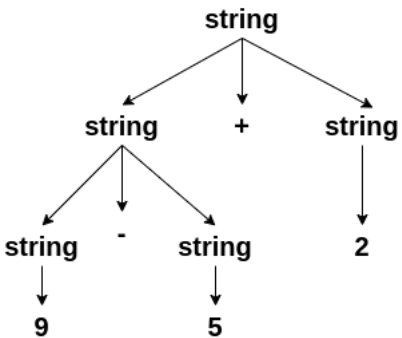

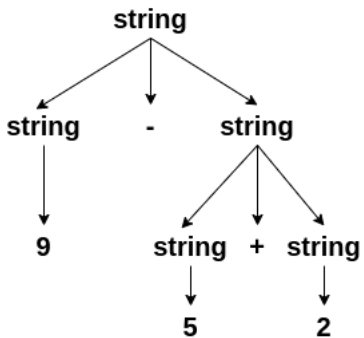

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  - **Bottom-up parsing:** Construction of the parse tree starts from the leaf nodes (tokens or terminals of the grammar) and proceeds towards root (start symbol). Ex YACC and BISON

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